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Counteracting Bone Mineral Loss and Secondary Sarcopenia in Inflammatory Bowel Disease Through Resistance and Weight-Bearing Exercise: A Narrative Review

Anna Hajduk, ORCID <https://orcid.org/0009-0003-1570-0608>

hajduk.ania2001@gmail.com

Independent Researcher, Lodz, Poland

Piotr Szymański, ORCID <https://orcid.org/0009-0006-2891-640X>

peterson2001@tlen.pl

Independent Researcher, Lodz, Poland

Klaudia Borycka, ORCID <https://orcid.org/0009-0006-0006-7813>

K.Borycka@interia.pl

Institute of Dentistry of the Central Clinical Hospital of the Medical University of Lodz,

ul. Pomorska 251, 92-213 Łódź, Poland

Aleksandra Szymańska, ORCID <https://orcid.org/0009-0002-2122-7522>

ola.szymanska9000@tlen.pl

Independent Researcher, Lodz, Poland

Filip Kostanek, ORCID <https://orcid.org/0009-0003-0243-5219>

filipkostanek@gmail.com

Independent Researcher, Lodz, Poland

Jagoda Stańska, ORCID <https://orcid.org/0009-0000-4861-2856>

lek.dent.jagoda.stanska@gmail.com

Institute of Dentistry of the Central Clinical Hospital of the Medical University of Lodz,
ul. Pomorska 251, 92-213 Łódź, Poland

Aleksandra Baranowska, ORCID <https://orcid.org/0009-0003-0373-7577>

olabaranowska200@gmail.com

Institute of Dentistry of the Central Clinical Hospital of the Medical University of Lodz,
ul. Pomorska 251, 92-213 Łódź, Poland

Aleksandra Dobrzańska, ORCID <https://orcid.org/0009-0002-4694-793X>

aleksandra.dobrzanska89@gmail.com

Institute of Dentistry of the Central Clinical Hospital of the Medical University of Lodz,
ul. Pomorska 251, 92-213 Łódź, Poland

Kacper Kmiecik, ORCID <https://orcid.org/0009-0007-2703-2538>

kacper.kmiecik@o2.pl

Institute of Dentistry of the Central Clinical Hospital of the Medical University of Lodz,
ul. Pomorska 251, 92-213 Łódź, Poland

Maria Berek, ORCID <https://orcid.org/0009-0006-2613-1123>

mariaberek12@gmail.com

Institute of Dentistry of the Central Clinical Hospital of the Medical University of Lodz,
ul. Pomorska 251, 92-213 Łódź, Poland

Corresponding Author

Anna Hajduk, hajduk.ania2001@gmail.com

Abstract

Background: Inflammatory bowel disease (IBD), including Crohn's disease and ulcerative colitis, is a chronic immune-mediated disorder of the gastrointestinal tract. It is associated with an increased risk of secondary osteoporosis and low bone mineral density (BMD), mainly due to chronic inflammation, nutrient malabsorption, long-term glucocorticoid therapy, and physical inactivity, which also contributes to sarcopenia.

Aim: To evaluate the evidence on resistance and weight-bearing exercise as non-pharmacological strategies to improve BMD, preserve muscle mass, and reduce secondary osteosarcopenia in patients with IBD.

Material and methods: A literature review was conducted using PubMed and Scopus. Randomized controlled trials and observational studies evaluating structured exercise interventions lasting at least 4 weeks in pediatric and adult IBD patients were included.

Results: Available evidence indicates that structured exercise, particularly combined resistance and weight-bearing training, improves lumbar spine and hip BMD. Studies in pediatric and young adult patients show that 6-12 months of training increases lean body mass and bone mineral content. Mechanical loading during resistance exercise stimulates bone formation and the release of anti-inflammatory myokines, including irisin, which may regulate the RANKL/OPG pathway and reduce inflammatory signaling. Exercise is generally safe, does not increase disease activity, and improves health-related quality of life.

Conclusions: Regular, individualized resistance and weight-bearing exercise should be considered an important adjunctive therapy for preventing bone loss and secondary sarcopenia

in IBD. Effective implementation requires collaboration between gastroenterologists and rehabilitation specialists to provide safe, tailored exercise programs.

Key words: inflammatory bowel disease; Crohn's disease; ulcerative colitis; resistance exercise; bone mineral density; osteosarcopenia

1. Introduction

Inflammatory bowel disease (IBD), including Crohn's disease (CD) and ulcerative colitis (UC), is a group of chronic, relapsing-remitting inflammatory disorders of the gastrointestinal tract (Trivić et al. 2023). Beyond intestinal symptoms, extraintestinal manifestations affect up to 50% of patients, with musculoskeletal deficits being among the most common and debilitating (Skrzypczak et al. 2021). IBD patients are at a significantly increased risk of secondary osteoporosis and osteopenia, with a pooled prevalence ranging from 20% to 50% (Ratajczak et al. 2022; Kreienbuehl et al. 2024). This skeletal deterioration is associated with a 38% higher overall risk of fractures compared to the general population, which severely impairs health-related quality of life (HRQoL) and increases clinical burden (Kreienbuehl et al. 2024; Ratajczak-Pawłowska et al. 2024).

The pathogenesis of bone loss in IBD is complex and multifactorial. Chronic systemic inflammation is a primary driver, as pro-inflammatory cytokines such as TNF- α , IL-1, IL-6, and IL-17 interfere with bone metabolism by directly activating osteoclasts and inhibiting osteoblast activity (Steell et al. 2021; Chang et al. 2022). Furthermore, long-term glucocorticoid therapy remains a major risk factor; glucocorticoids induce osteoblast and osteocyte apoptosis while decreasing intestinal calcium absorption (Hofbauer et al. 2025; Steell et al. 2021). Nutritional deficiencies, particularly the malabsorption of calcium, magnesium, and vitamin D due to intestinal damage or surgical resections, further compromise bone mineralization (Ratajczak et al. 2022; Skrzypczak et al. 2021). Additionally, the co-occurrence of sarcopenia (loss of muscle mass and strength) in IBD patients creates a vicious cycle where the reduced mechanical loading generated by skeletal muscles accelerates bone resorption (Skrzypczak et al. 2021; Trivić et al. 2023).

Given these risks, non-pharmacological interventions like structured exercise are increasingly recognized as important adjunctive therapies. Resistance training and weight-bearing exercises

provide an osteogenic stimulus through mechanotransduction (Chang et al. 2022; Jones et al. 2022). In this process, mechanical forces from gravity and muscle contractions are sensed by osteocytes and converted into biological signals, such as the activation of the Wnt/ β -catenin pathway, which promotes bone formation and enhances bone microarchitecture (Chang et al. 2022; Zerlotin et al. 2022). Moreover, exercise influences the "muscle-bone unit" by triggering the release of anti-inflammatory myokines, such as irisin, which may suppress systemic inflammation and modulate the RANKL/OPG signaling axis to slow bone loss (Chang et al. 2022; Zerlotin et al. 2022; Müller et al. 2026). Physical activity also helps reduce visceral fat and improve cardiovascular health, contributing to improved metabolic health in IBD (Ata and Eyigor 2024; Müller et al. 2026).

Research Objective

The main objective of this narrative review is to evaluate the impact of structured resistance training and weight-bearing exercise on bone mineral density (BMD), muscle mass, and the management of osteopenia and secondary sarcopenia in pediatric and adult patients with inflammatory bowel disease.

Research Problems

1. What are the physiological and molecular mechanisms, such as mechanotransduction and myokine secretion, through which mechanical loading influences bone remodeling in the context of chronic IBD-related inflammation?
2. To what extent do specific exercise modalities, primarily combined resistance and high-impact loading, improve site-specific areal and volumetric BMD at the lumbar spine and hip in IBD populations?
3. Are structured exercise programs safe and well-tolerated for IBD patients across different levels of disease activity, and do they pose a risk for disease flares?

Research Hypotheses

1. Structured resistance training and weight-bearing exercises significantly improve bone mineral density and content in IBD patients by providing an osteogenic stimulus that counteracts systemic bone loss (Trivić et al. 2023; Chang et al. 2022).

2. Regular mechanical loading helps modulate the systemic inflammatory environment in IBD by promoting anti-inflammatory myokine release, thereby slowing the rate of osteoclast-mediated bone resorption (Müller et al. 2026; Zerlotin et al. 2022).
3. Personalized, structured exercise programs are safe for patients in clinical remission or with mild disease activity and do not exacerbate gastrointestinal symptoms or trigger disease relapses (Trivić et al. 2023; Müller et al. 2026).

2. Research materials and methods

This comprehensive narrative review provides a qualitative synthesis of the available scientific literature. The methodology follows established frameworks for identifying, synthesizing, and critically analyzing evidence regarding the impact of physical activity on musculoskeletal health in clinical populations.

2.1. Characteristics of the Study Populations

The study populations described in the reviewed literature encompass a wide clinical spectrum of patients diagnosed with inflammatory bowel disease (IBD), including both Crohn's disease (CD) and ulcerative colitis (UC) (Trivić et al. 2023; Kasznár et al. 2026). The age range of participants is extensive, covering pediatric and adolescent cohorts (aged 10–18 years), young adults with childhood-onset disease, and middle-aged to elderly populations (Ahn and Yoo 2023; Sigurdsson et al. 2021; Ratajczak-Pawłowska et al. 2024).

Clinical status varies across the studies, with many investigations focusing on patients in stable clinical remission – defined by standardized indices such as the Pediatric Crohn's Disease Activity Index (PCDAI) or the Harvey-Bradshaw Index (HBI) – while others include those with mild-to-moderate disease activity (Jones et al. 2020; Trivić et al. 2023; Li et al. 2026). A high prevalence of compromised bone health is noted at baseline, with significant proportions of patients presenting with osteopenia, secondary sarcopenia, or osteoporosis, often exacerbated by a history of glucocorticoid therapy and chronic systemic inflammation (Ratajczak et al. 2022; Lewandowski et al. 2023; Kreienbuehl et al. 2024).

2.2. Assessment Methods and Exercise Interventions

The procedures identified in the literature utilize standardized clinical assessments. Bone mineral density (BMD) is predominantly measured using dual-energy X-ray absorptiometry

(DXA), which represents the gold standard for evaluating areal BMD (aBMD) at the lumbar spine, femoral neck, and total body (Aibar-Almazán et al. 2022; Dai et al. 2023). Diagnostic metrics include T-scores, Z-scores, and Bone Mineral Content (BMC) (Ratajczak-Pawłowska et al. 2024). Some advanced studies employ high-resolution peripheral quantitative computed tomography (HR-pQCT) to assess volumetric BMD (vBMD) and cortical/trabecular microarchitecture (Steell et al. 2021; Sigurdsson et al. 2023). Muscle mass and body composition are evaluated via DXA-derived lean body mass (LBM) and the Skeletal Muscle Index (SMI).

In addition to imaging, bone metabolism is monitored via serum biomarkers of bone turnover, specifically C-terminal telopeptide of type 1 collagen (CTX) and osteocalcin, alongside assessments of 25-hydroxyvitamin D [25(OH)D] and parathyroid hormone (PTH) levels (Lewandowski et al. 2023; Ratajczak et al. 2022).

Exercise protocols described in the analyzed literature fall into three primary categories:

- **Structured resistance and impact training:** High-effort resistance movements and weight-bearing exercises (e.g., multidirectional jumps, elastic band resistance, free weights), typically performed 2–3 times weekly (Jones et al. 2020; Jones et al. 2022).
- **Bodyweight high-impact programs:** Home-based protocols involving progressive bodyweight exercises such as squats, lunges, planks, and push-ups (Trivić et al. 2023).
- **Low-impact and aerobic modalities:** Progressive walking programs, swimming, cycling, and yoga interventions (Robinson et al. 1998; Tew et al. 2019; Kasznár et al. 2026).

2.3. Literature Search and Synthesis Strategy

Literature searches were conducted across electronic databases, including PubMed and Scopus. The search strategy employed specific keywords, MeSH terms, and Boolean operators, including combinations of: "inflammatory bowel disease", "Crohn's disease", "ulcerative colitis", "resistance training", "weight-bearing exercise", "bone mineral density", "osteoporosis", "secondary sarcopenia", and "muscle-bone axis".

The synthesis included publications spanning from the late 1990s to recent literature to capture both foundational longitudinal data and recent clinical advancements. Eligible study designs comprised randomized controlled trials (RCTs), non-randomized interventional trials,

prospective observational cohorts, and systematic reviews with meta-analyses evaluating exercise interventions lasting 4 or more weeks.

2.3.1. AI Declaration

Artificial intelligence tools were used strictly in an auxiliary capacity to assist with language editing, improving English grammar, readability, and stylistic precision. All conceptual design, data evaluation, synthesis of evidence, and formulation of conclusions were performed independently by the authors, who retain full responsibility for the manuscript.

2.3.2. Statistical Methods Reported in Included Studies

As a narrative review, this study involved no primary statistical calculations or raw data collection; findings were derived from the synthesis of reported results in the primary literature. Original studies employed diverse analytical approaches, ranging from descriptive statistics (means $\pm$ SD or medians with interquartile ranges) and comparative tests (Student's t-tests, ANOVA, Mann-Whitney U) to evaluate group differences (Trivić et al. 2023; Ratajczak-Pawłowska et al. 2024). Multivariable linear and logistic regression models were frequently applied to adjust for confounding variables such as corticosteroid dose and disease duration (Sigurdsson et al. 2021; Ahn and Yoo 2023). Meta-analyses in the source literature utilized random-effects models expressing effect sizes as mean differences (MD) or standardized mean differences (SMD) with 95% confidence intervals (95% CI) (Kasznár et al. 2026; Li et al. 2026). Additionally, recent works utilized two-sample Mendelian randomization to assess genetically predicted causal associations (Dai et al. 2023; Wu et al. 2023). In majority of included studies, statistical significance was defined as $p < 0.05$.

3. Research Results

3.1. Impact of Resistance Training on Bone Mineral Density (BMD) and Bone Microarchitecture

Clinical evidence synthesized from the included studies indicates that structured resistance and weight-bearing exercises serve as a powerful osteogenic stimulus, although the magnitude of skeletal adaptation varies by age, compliance, and specific anatomical site. These interventional

findings indicate that mechanical loading through structured exercise may counteract systemic bone loss and improve skeletal outcomes in patients with IBD.

In pediatric and adolescent IBD populations, a 6-month home-based high-impact bodyweight program led to a statistically significant increase in total body less head (TBLH) BMD, rising from $0.959 \pm 0.023 \text{ g/cm}^2$ to $0.988 \pm 0.025 \text{ g/cm}^2$ ($p < 0.001$) (Trivić et al. 2023). This clinical improvement was mirrored by a significant rise in age- and sex-adjusted BMD Z-scores from -0.35 ± 0.15 to -0.28 ± 0.17 ($p = 0.020$) (Trivić et al. 2023). Observational data on young adults with childhood-onset disease indicate that those engaging in high levels of physical exercise (≥ 4 hours/week) can reach total body BMD Z-scores comparable to healthy normative controls, whereas sedentary patients maintain significantly lower Z-scores (-1.08 vs. -0.25 , $p = 0.002$) (Sigurdsson et al. 2021).

In the adult IBD population, site-specific responses to mechanical loading are evident, particularly at the lumbar spine and hip. A 6-month supervised program of combined impact and resistance training (three times weekly) achieved significant gains in lumbar spine areal BMD (adjusted mean difference: 0.036 g/cm^2 ; 95% CI: $0.024\text{--}0.048$; $p < 0.001$) compared to usual care (Jones et al. 2020; González-Aroca et al. 2026). While site-specific improvements at the femoral neck are often more modest, subgroup analyses of patients completing ≥ 10 sessions per month revealed a statistically significant 4.67% increase in greater trochanter aBMD ($p = 0.02$) (Robinson et al. 1998; González-Aroca et al. 2026).

Advanced imaging techniques such as high-resolution peripheral quantitative computed tomography (HR-pQCT) provide deeper insight into bone quality beyond simple density. Research indicates that young adult males with IBD exhibit significant structural deficits, including smaller cortical area (126.1 mm^2 vs. 151.1 mm^2), lower cortical volumetric BMD (854.4 mg/cm^3 vs. 878.5 mg/cm^3), and thinner trabeculae (0.084 mm vs. 0.089 mm) compared to healthy controls (Sigurdsson et al. 2023). Importantly, multivariable regression models demonstrate that the association between physical exercise and these microarchitectural parameters is largely mediated by the skeletal muscle index (SMI), suggesting that exercise-induced muscle development is the primary driver of bone structural maintenance (Steell et al. 2021; Sigurdsson et al. 2023).

3.2. Biochemical Bone Markers and Turnover Dynamics in IBD Patients

The impact of regular exercise extends to the biochemical level, effectively shifting bone turnover dynamics toward formation. The synthesized literature suggests that physical activity may beneficially influence bone remodeling markers, even in the presence of underlying inflammatory stress.

In clinical cohorts, sedentary IBD patients presenting with osteoporosis show significantly higher serum concentrations of C-terminal telopeptide of type 1 collagen (CTX), a marker of bone resorption, compared to their physically active counterparts (Lewandowski et al. 2023). Conversely, active patients demonstrate higher levels of osteocalcin, a primary marker of bone formation (Lewandowski et al. 2023). These findings suggest that frequent physical activity acts as a modifiable protective factor by normalizing the rate of bone remodeling (Ratajczak et al. 2022; Lewandowski et al. 2023).

Mechanistically, the RANK/RANKL/OPG signaling axis represents a central pathway through which exercise modulates bone metabolism. Chronic inflammation in IBD typically upregulates RANKL, which binds to RANK on osteoclast precursors to trigger bone resorption (Chang et al. 2022; Kreienbuehl et al. 2024). Mechanical loading during resistance training and impact exercise has been shown to upregulate osteoprotegerin (OPG), a decoy receptor that binds to RANKL, thereby inhibiting osteoclast maturation and survival (Chang et al. 2022; Zerlotin et al. 2022). Furthermore, vibration and impact loading stimulate osteoblasts to increase the production of P1NP (procollagen type 1 N-terminal propeptide), reinforcing the anabolic phase of the bone cycle (Chang et al. 2022; Lewandowski et al. 2023).

3.3. Inflammatory Cytokine Modulation and the Muscle-Bone Crosstalk

Structured exercise may facilitate a beneficial bidirectional interaction between muscle and bone through the secretion of exercise-induced myokines. Muscle contractions during resistance training stimulate the release of irisin, which acts as a vital signaling mediator within the muscle-bone unit (Chang et al. 2022; Zerlotin et al. 2022). Irisin has been reported to suppress the NF- κ B signaling pathway, effectively inhibiting the expression of pro-inflammatory cytokines such as TNF- α , IL-1, and IL-6 – the primary molecular drivers of systemic bone loss in IBD (Chang et al. 2022; Sobieszek and Idzik 2025).

Meta-analyses of exercise interventions in IBD confirm a trend toward systemic inflammatory reduction, with reported decreases in C-reactive protein (CRP) (MD: -0.13 mg/L) and erythrocyte sedimentation rate (ESR) (MD: -6.21 mm/h) (Kasznár et al. 2026). Furthermore, resistance training significantly enhances lean body mass (LBM), with mean gains of 1.6 kg in pediatric cohorts and 1.59 kg in adults (Cronin et al. 2019; Trivić et al. 2023). The increase in skeletal muscle mass index (SMI) Z-scores is strongly correlated with femoral neck BMD, reinforcing the concept that the mechanical pull of developing muscle tissue provides the necessary strain to activate the Wnt/ β -catenin and Piezo1 mechanosensors in osteocytes, leading to bone mineralization (Sigurdsson et al. 2021; Chang et al. 2022; Yang and Jeon 2024).

3.4. Impact of Pharmacotherapy and Corticosteroid Co-interventions on Exercise Adaptations

The effectiveness of exercise in IBD is often influenced by the patient's pharmacological background, particularly long-term glucocorticoid (GC) use. GCs are known to induce osteoblast and osteocyte apoptosis while simultaneously increasing myostatin expression, which leads to muscle atrophy and decreased BMD (Skrzypczak et al. 2021; Hofbauer et al. 2025). Despite these catabolic pressures, interventional data show that exercise remains beneficial; for instance, in the pediatric cohort studied by Trivić et al. (2023), the cumulative dose of glucocorticosteroids was not associated with an inability to gain BMD or LBM, suggesting that structured training can counteract steroid-induced deficits.

Conversely, biological therapies, such as anti-TNF agents (e.g., infliximab), appear to have a synergistic effect when combined with exercise. These therapies reduce the systemic inflammatory burden by neutralizing TNF- α , which allows for more efficient bone formation and muscle hypertrophy (Ata and Eyigor 2024; Kreienbuehl et al. 2024). Studies have shown that maintenance treatment with infliximab is associated with improved lumbar spine BMD over 6–12 months, and adding structured resistance training can further enhance these gains by stabilizing the muscle-bone unit (Steell et al. 2021; Yang and Jeon 2024).

3.5. Safety, Gastrointestinal Tolerance, Disease Activity, and Quality of Life (HRQoL)

Available evidence indicates that resistance and impact training are generally safe and well tolerated among IBD patients in clinical remission or with mild disease activity. Meta-analytic data indicate a significant improvement in clinical disease activity following structured exercise

programs, with a standardized mean difference (SMD) of -0.46 ($p = 0.01$) (Jones et al. 2022). Critically, structured interventions do not cause significant increases in fecal calprotectin levels or trigger disease flares (Tew et al. 2019; Seeger et al. 2020). Patients with severe disease, however, remain underrepresented in the literature, often avoiding exercise due to fear of increased toilet urgency or abdominal pain (Sinclair et al. 2024).

The psychological impact of exercise is substantial. Meta-analyses show that physical activity significantly improves health-related quality of life (HRQoL) (SMD: 0.55), with the most robust gains found in the systemic subscale, which encompasses improvements in fatigue, sleep quality, and energy levels (Jones et al. 2022; Kasznár et al. 2026). In a French multicenter study, physical activity was positively associated with acceptance of the disease and overall QoL across all levels of symptom intensity (Derbey et al. 2024). While minor side effects like transient musculoskeletal soreness or light-headedness have been reported, no serious adverse events were noted across trials (Jones et al. 2022).

3.6. Dosing, Load Progression, and Adherence Rates in Clinical Exercise Protocols

Optimal exercise protocols for bone health in IBD generally involve a frequency of three sessions per week and a duration of 12 to 26 weeks (Jones et al. 2022; Kasznár et al. 2026). Intensities for resistance training typically range from 60% to 70% of 1-repetition maximum (1RM), focusing on multi-joint movements and progressive loading (Candow et al. 2002; Jones et al. 2020). Impact loading often consists of multidirectional jumps or skipping, which provides the high-strain rates necessary to trigger bone adaptation (Jones et al. 2020).

Adherence rates in supervised and tailored programs are high. Pediatric home-based protocols achieved 86% adherence (participants completing $> 75\%$ of sessions) through weekly telephone support (Trivić et al. 2023). Adult studies also show strong feasibility; for instance, supervised programs that gradually taper toward self-management report sustained engagement (Tew et al. 2019; Jones et al. 2020). The primary barriers to adherence remain asthenia (fatigue) and the lack of disease-specific guidance from healthcare providers, highlighting the need for multidisciplinary rehabilitation teams (Ata and Eyigor 2024; Derby et al. 2024).

Table 1. Comprehensive Synthesis of Literature Findings

Study / Authors (Year)	Study Design & Population (n, CD/UC, remission status)	Primary Outcomes (BMD / Bone Markers / Cytokines)	Key Results & Clinical Findings
Robinson et al. (1998)	RCT; n=117 CD; Remission to mildly active.	DXA aBMD (Spine, Hip); Fracture risk.	↑ 4.67% Trochanter aBMD in compliant subgroup; Safe long-term intervention.
Jones et al. (2020)	RCT; n=47 CD; Stable remission/inactive.	DXA aBMD (Spine, Hip); Fecal calprotectin.	↑ Lumbar BMD (0.036 g/cm ²); ↑ Strength; Stable GI inflammation.
Trivić et al. (2023)	Pre-post; n=42 pediatric CD/UC; Clinical remission.	DXA BMD; Z-score; LBM; FM; Muscle fitness.	↑ TBLH BMD; ↑ BMD Z-score (-0.35 to -0.28); ↑ LBM (1.6 kg); No adverse events.
Sigurdsson et al. (2021)	Cross-sectional; n=72 young adult IBD; Long-standing disease.	DXA BMD Z-scores; SMI; Fat %.	High PE (>4h/w) reaches normal total BMD; Sedentary patients have severe deficits.
Cronin et al. (2019)	RCT Crossover; n=17 IBD; Remission.	DXA LBM; IL-6; TNF-α; IL-10.	↑ LBM (1.59 kg); ↓ 2.1% body fat; No change in systemic cytokines.
Lewandowski et al. (2023)	Observational; n=232 IBD; Mixed activity.	Osteocalcin; CTX; PTH; Vitamin D.	Daily PE correlates with ↓ Osteoporosis risk; ↓ Resorption marker CTX; ↑ Formation marker Osteocalcin.
Kasznár et al. (2026)	Meta-analysis; n=498 IBD; mixed.	HRQoL; Disease activity; VO2max.	Significant ↑ HRQoL (SMD 0.55); Non-significant trend in ↓ CRP and calprotectin.

Table 2. Exercise Prescription Parameters and Load Characteristics in Included Studies

Study / Authors (Year)	Exercise Modality & Equipment	Intensity (% 1RM / RPE) & Frequency	Intervention Duration	Safety / Adverse Events Reported
Robinson et al. (1998)	Home-based floor exercises (trunk/leg); progressive weights.	Progressive load; 2 sessions/week.	12 months.	No flares or serious adverse events.
Jones et al. (2020)	Impact (skipping/jumps) + Resistance (TheraBands).	High-effort + resistance; 3 sessions/week.	6 months.	2 cases of transient light-headedness; no flares.

Trivić et al. (2023)	Home-based bodyweight (squats, planks, push-ups).	High-impact bodyweight; 3 sessions/week.	26 weeks.	No adverse events reported; no disease exacerbation.
Candow et al. (2002)	Resistance training machines.	60–70% 1RM; 3 sets of 8–10 reps; 3x/week.	12 weeks.	Not reported.
Tew et al. (2019)	Cycling (HIIT vs. MICT).	HIIT (90% W _{peak}) vs. MICT (35% W _{peak}); 3x/week.	12 weeks.	No change in CDAI or calprotectin.
Seeger et al. (2020)	Muscle training (12 exercises) vs. Endurance.	Moderate intensity; 3 sessions/week.	12 weeks.	Well-tolerated; no symptom worsening.

4. Discussion

The findings of this review underscore the potential of resistance training as a potent non-pharmacological intervention for mitigating the skeletal and muscular deficits inherent to inflammatory bowel disease (IBD). The biological pathways underlying bone adaptation in response to resistance training are closely related to the mechanisms described in the first research problem and are consistent with the hypothesis that exercise-induced mechanical loading may influence bone remodeling through molecular signaling pathways. These adaptations are primarily rooted in mechanotransduction, whereby osteocytes sense mechanical forces and convert them into biological signals regulating bone remodeling (Jones et al. 2022). Specifically, mechanical loading triggers stretch-activated ion channels and receptors such as Piezo1/2 and integrins, which activate the Wnt/ β -catenin signaling pathway, a key driver of osteoblast differentiation and bone formation (Chang et al. 2022). In the pathological environment of IBD, chronic systemic inflammation characterized by elevated pro-inflammatory cytokines like TNF- α and IL-6 disrupts this homeostasis by upregulating the expression of receptor activator of nuclear factor κ B ligand (RANKL) and suppressing osteoprotegerin (OPG), thereby shifting the balance toward excessive bone resorption (Steell et al. 2021; Kreienbuehl et al. 2024). Resistance training may counteract this imbalance by modulating the RANKL/OPG ratio and suppressing NF- κ B-mediated inflammatory pathways (Steell et al. 2021; Chang et al. 2022). Furthermore, evidence suggests that moderate-intensity exercise can downregulate the colonic expression of pro-inflammatory cytokines while increasing anti-inflammatory mediators like IL-10, thus restoring systemic immune homeostasis (Sobieszek and Idzik 2025).

The interaction between resistance training and IBD pharmacotherapy represents an important aspect of the mechanisms considered within the first research problem, particularly regarding the ability of exercise to mitigate treatment-related and inflammation-related musculoskeletal complications. Glucocorticoids, frequently utilized for managing disease flares, exert profound catabolic effects on bone and muscle by increasing osteocyte apoptosis, decreasing osteoblastogenesis, and stimulating myostatin expression (Skrzypczak et al. 2021; Hofbauer et al. 2025). High-intensity resistance training serves as an important supportive strategy in countermeasuring glucocorticoid-induced osteoporosis (GIOP), providing an osteogenic stimulus that mitigates bone mineral density (BMD) loss and muscle atrophy (Dent et al. 2023; Reddy et al. 2026). Conversely, biological therapies such as anti-TNF agents (e.g., infliximab) appear to have synergistic effects with exercise; by suppressing the underlying inflammatory burden, these medications reduce the systemic cytokine levels that would otherwise inhibit bone formation and drive muscle degradation (Steell et al. 2021; Kreienbuehl et al. 2024). However, the timing of exercise remains essential, as high-intensity or exhaustive exercise initiated during active flares might exacerbate intestinal injury and oxidative stress, suggesting a "U-shaped" response where moderate, progressive loading is most beneficial (Sobieszek and Idzik 2025; Kasznár et al. 2026).

The evidence regarding structural and density adaptations supports the hypothesis that resistance and weight-bearing exercise may contribute to the preservation of skeletal health in IBD through muscle-bone crosstalk. This interaction, mediated through biochemical signaling between these tissues, is central to preventing secondary sarcopenia and maintaining bone integrity.. Skeletal muscle acts as an endocrine organ, secreting myokines such as irisin and IL-15 that directly influence bone metabolism (Skrzypczak et al. 2021). Irisin, produced following muscle contraction, may promote osteoblast differentiation via the Wnt/ β -catenin pathway and promotes recovery of bone turnover by reducing TNF- α and RANKL (Zerlotin et al. 2022). In IBD patients, a vicious cycle of osteosarcopenia exists where inflammation promotes both low muscle mass (sarcopenia) and low BMD (osteoporosis), which are independent predictors of surgical failure and poor quality of life (Skrzypczak et al. 2021; Yang and Jeon 2025). Resistance training effectively disrupts this cycle by inducing skeletal muscle hypertrophy and enhancing cardiorespiratory fitness, which in turn improves bone mineral apparent density and skeletal microarchitecture (Jones et al. 2022; Trivić et al. 2023). Furthermore, muscle mass serves as a vital mediator for bone health in IBD, as indicated by findings that increased skeletal

muscle index (SMI) is strongly associated with improved volumetric BMD in the weight-bearing femoral neck (Sigurdsson et al. 2021; Sigurdsson et al. 2023).

The evidence reviewed in this study supports the hypothesis that appropriately prescribed exercise programs are safe and well tolerated in selected IBD populations. Clinical trials in both pediatric and adult populations have indicated that supervised, structured exercise programs are safe and do not aggravate IBD symptoms or trigger disease relapses (Trivić et al. 2023; Kasznár et al. 2026). In fact, such interventions significantly improve health-related quality of life, particularly regarding fatigue and sleep quality (Derbey et al. 2024; Kasznár et al. 2026). However, many IBD patients remain physically inactive due to fears of diarrhea, abdominal pain, and a lack of guidance from healthcare providers (Ata and Eyigor 2024; Sinclair et al. 2024). Multidisciplinary management involving physiotherapists is essential to develop individualized programs that incorporate progressive resistance training and weight-bearing impact activities, ensuring that mechanical loading is sufficient to elicit an osteogenic effect (Jones et al. 2022; Ata and Eyigor 2024). Clinicians must also prioritize the screening of musculoskeletal health using dual-energy X-ray absorptiometry (DXA), especially for patients on long-term corticosteroids or those with low body mass index (Kreienbuehl et al. 2024; Reddy et al. 2026). Combining exercise with optimal nutrition, including adequate vitamin D and calcium intake, is paramount for maximizing skeletal gains (Steell et al. 2021; Dent et al. 2023).

Despite the evidence presented, several methodological limitations in the current literature must be addressed. Most randomized controlled trials have focused exclusively on patients with Crohn's disease in clinical remission, resulting in a potential selection bias and a lack of data regarding ulcerative colitis and patients with active disease states (González-Aroca et al. 2026). Furthermore, variations in exercise modalities, training durations, and reporting of outcomes – often omitting absolute BMD changes in favor of percentage changes – limit the ability to conduct robust quantitative meta-analyses (Jones et al. 2022; González-Aroca et al. 2026). Future research should employ larger, multicenter clinical trials that include diverse IBD phenotypes and utilize high-resolution imaging such as high-resolution peripheral quantitative computed tomography (HR-pQCT) to assess changes in trabecular and cortical microarchitecture (Sigurdsson et al. 2023; González-Aroca et al. 2026). Studies should also aim to define optimal exercise dose-response relationships and explore the long-term sustainability of musculoskeletal improvements (Jones et al. 2022; Sobieszek and Idzik 2025). Establishing specific guidelines for the intensity, frequency, and type of resistance training will be critical

for the effective integration of exercise as a standard component of holistic IBD management (Ata and Eyigor 2024; Sinclair et al. 2024).

5. Conclusions

The comprehensive synthesis of evidence presented in this review indicates that structured resistance training and weight-bearing impact exercise may represent an effective non-pharmacological strategy for counteracting systemic bone mineral density (BMD) loss and sarcopenia in inflammatory bowel disease. Progressive mechanical loading significantly enhances volumetric BMD and skeletal muscle index, potentially slowing the pathological progression of osteosarcopenia without triggering gastrointestinal flares or exacerbating disease activity (Sigurdsson et al. 2021; Trivić et al. 2023; Kasznár et al. 2026).

At the biological level, the efficacy of resistance training is mediated by mechanotransduction, which activates anabolic signaling pathways while modulating the systemic inflammatory burden. Mechanical strain has been associated with modulation of inflammatory pathways, including changes in TNF- α and IL-6 signaling, while improving the RANKL/OPG ratio and stimulating myokine-mediated bone formation via irisin release (Chang et al. 2022). This biological modulation may support protection against glucocorticoid-induced osteoporosis, mitigating the severe catabolic effects of long-term steroid therapy on bone and muscle tissue (Dent et al. 2023; Hofbauer et al. 2025).

The available evidence support the integration of structured, individually tailored strength training as an important component of holistic IBD management. Tailored resistance exercise is safe, well-tolerated, and significantly enhances health-related quality of life and reduces fatigue in patients with quiescent or mild disease (Derbey et al. 2024; Kasznár et al. 2026). Multidisciplinary care teams should consider prioritizing early musculoskeletal screening using dual-energy X-ray absorptiometry (DXA) and combine tailored mechanical loading protocols with optimal nutritional support, including vitamin D and calcium, to maximize long-term skeletal health and overall clinical outcomes (Ata and Eyigor 2024; Reddy et al. 2026).

Disclosure

Supplementary Materials

Not applicable

Author Contributions

All authors have read and agreed to the published version of the manuscript.

Conceptualization: Anna Hajduk, Piotr Szymański, Klaudia Borycka

Methodology: Aleksandra Szymańska, Filip Kostanek, Maria Berek

Investigation: Anna Hajduk, Aleksandra Baranowska, Jagoda Stańska

Formal analysis: Aleksandra Szymańska, Filip Kostanek, Kacper Kmieciak

Writing - original draft preparation: Klaudia Borycka, Aleksandra Dobrzańska, Anna Hajduk

Data curation: Piotr Szymański, Aleksandra Baranowska, Jagoda Stańska

Visualization: Anna Hajduk, Aleksandra Dobrzańska

Project administration: Anna Hajduk, Maria Berek, Kacper Kmieciak

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Conflicts of Interest

The authors deny any conflict of interest.

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